# CHE 321 – Chemical Reaction Engineering
# Fall 2019

## Home Work – 4 (20 points)

## Due: 12 noon, 4th Oct 2019

## Problem 1:

Consider the cracking reactions of ethane. They can be modelled as I order reactions. The reactions are given by

$$C_2H_6 \leftrightharpoons C_2H_4 + H_2 \qquad 1$$

$$C_2H_4 \leftrightharpoons C_2H_2 + H_2 \qquad 2$$

$$C_2H_2 + 3H_2 \leftrightharpoons C_2H_6 \qquad 3$$

Let $C_2H_6$, $C_2H_4$ and $C_2H_2$ be denoted by A, B and C. Neglect hydrogen during the analysis.

The rate equations are given as,

$$r_1 = k_{1f} X_A - k_{1b} X_B$$

$$r_2 = k_{2f} X_B - k_{2b} X_C$$

$$r_3 = k_{3f} X_c - k_{3b} X_A$$

Here X denotes the mole fraction and the subscripts f and b indicate forward and backward reactions. The experimental data is provided in the Excel sheet attached. Estimate the rate constant matrix for this reaction system.

## Procedure

1) Write the mole balance equations for species A, B and C in terms of mole fractions.
2) Use equation 129 from the class notes to estimate the rate constants.
3) Look at the data sheet provided. Plot $\ln(|X_j - X_j^*|)$ Vs t for j = A, B, C. The slopes and intercepts will give eigen values and $\ln(\alpha_j Z_j)$.
4) Estimate $\lambda_3$ and $Z_3$ from the experiment 1 data and $\lambda_2$ and $Z_2$ from the experiment 2 data. $\lambda_1 = 0$ and $Z_1 = [-0.5866\ -0.6433\ -0.4920]^T$

5) $\alpha_3 = 0.0215$ and $\alpha_2 = 0.9$. Take the average values of $\lambda$ from experiment 1 to get $\lambda_3$. Similarly take the average values of $\lambda$ from experiment 2 to get $\lambda_2$.
6) Find the k matrix which is given by $Z^{-1}\Lambda Z$ where Z is the matrix formed by taking the eigen vectors as the columns and $\Lambda$ is the diagonal matrix formed by taking the eigen values as the diagonal elements.

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## Problem 2:

1) Email the programs as M. file to your assigned TA.
2) 5 points will be deducted for submissions after the deadline.
3) Provide the complete code along with the functions for full credits. Mention all the assumptions and explanations wherever necessary.
4) Submit the homework in a zip file. All the figures and explanation should be in a pdf CHE321_HW6_LastName_FirstName.pdf.
5) In the case that the submitted file is saved under a name other than that of the student, 3 points will be deducted.

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1. Consider a biochemical reactor operated in the batch mode with initial concentration of substrate at 10 gm/liter and biomass at 0.1 gm/liter. The growth rate of the cells is assumed to follow Monod kinetics given by

$$\alpha S \longrightarrow B$$

$$\frac{\mu c_s c}{K_s + c_s}$$

where $\mu$ is the maximum growth rate, $K_s$ is the Michaelis constant, $c$ is the concentration of biomass, and $c_s$ is the concentration of the substrate. The constants in the above rate expression are unknown. The extent of growth $\xi$, is reported from experiments as below.

| Time (hours) | Biomass concentration gm/liter | Substrate concentration gm/liter |
|---|---|---|
| 0 | 0.1 | 10 |
| 0.5 | 0.17 | 10.17 |
| 1.0 | 0.26 | 9.38 |
| 1.5 | 0.40 | 8.68 |
| 2.0 | 0.77 | 9.48 |
| 2.5 | 1.19 | 8.12 |
| 3.0 | 1.84 | 6.31 |

| | | |
|---|---|---|
| 3.5 | 3.08 | 4.15 |
| 3.7 | 3.57 | 2.76 |
| 3.8 | 4.15 | 2.20 |
| 3.9 | 4.32 | 1.36 |
| 4.0 | 4.88 | 0.64 |
| 4.5 | 4.89 | 0.01 |
| 5.0 | 5 | 0 |

Use Bellman's method of quasilinearization to estimate the parameters, $\mu$, $K_s$ from the above data.

- Calculate α and ξ using biomass and substrate concentrations in your main function.
- Define : xguess=[zeros(1,length(Time)) ; $10^{-5}$ *ones(1, length(time)); $10^{-5}$ *ones(1, length(time))] in your main function.

- Define Jacobian, f, P, H as functions in ***your main function***.

$$F(x) = [\ r;0;0] = [\frac{x(2)(C_{s0}+\alpha x(1))(C_0+x(1))}{(x(3)+C_{s0}+\alpha x(1)}; 0; 0]$$

$$J(x) = \begin{bmatrix} \frac{\partial F(1.1)}{\partial x(1)} & \frac{\partial F(1.1)}{\partial x(2)} & \frac{\partial F(1.1)}{\partial x(3)} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

P(J,F,x,time) = $e^{-J(1.1)*time} * (J(1.1) - F(1.:) * x) * time$

Function dh = H(t,h0)

dh = J(xold)*h0

- We are using a matlab function called fminunc to minimize the error to obtain x0.

  x0 = fminunc(@errorfunc,xguess)

- Define the  Error function in your program,

  Function E = errfunc(x0)

  xold = xnew

  H1 = ode45(@H,time,[1,0,0])

  H2 = ode45(@H,time,[0,1,0])

  H3 = ode45(@H,time,[0,0,1])

  Compute P

  Xnew = P+ H1*x0+H2*x0+h3*x0 for every value of time.

  E = $\sum(x_{new} - \xi)^2$

- Create a while loop in your main function

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While $(\sum(x_{new} - \xi)^2 > 0.01)$

x0 = fminunc(@errorfunc,xguess)

H1 = ode45(@H,time,[1,0,0])

H2 =o de45(@H,time,[0,1,0])

H3 = ode45(@H,time,[0,0,1])

Compute P

xnew = P+ H1*x0+H2*x0+h3*x0 for every value of time.

End while

*Hint:xguess = [ $\xi$(time) ;$\mu$;$K_s$] for every value of Time.*

*Hint:xold is equal to x at a particular time, so xold is a vector.*

*Hint: the input of F and J and P is always xold..*

Use the chart below for better understanding

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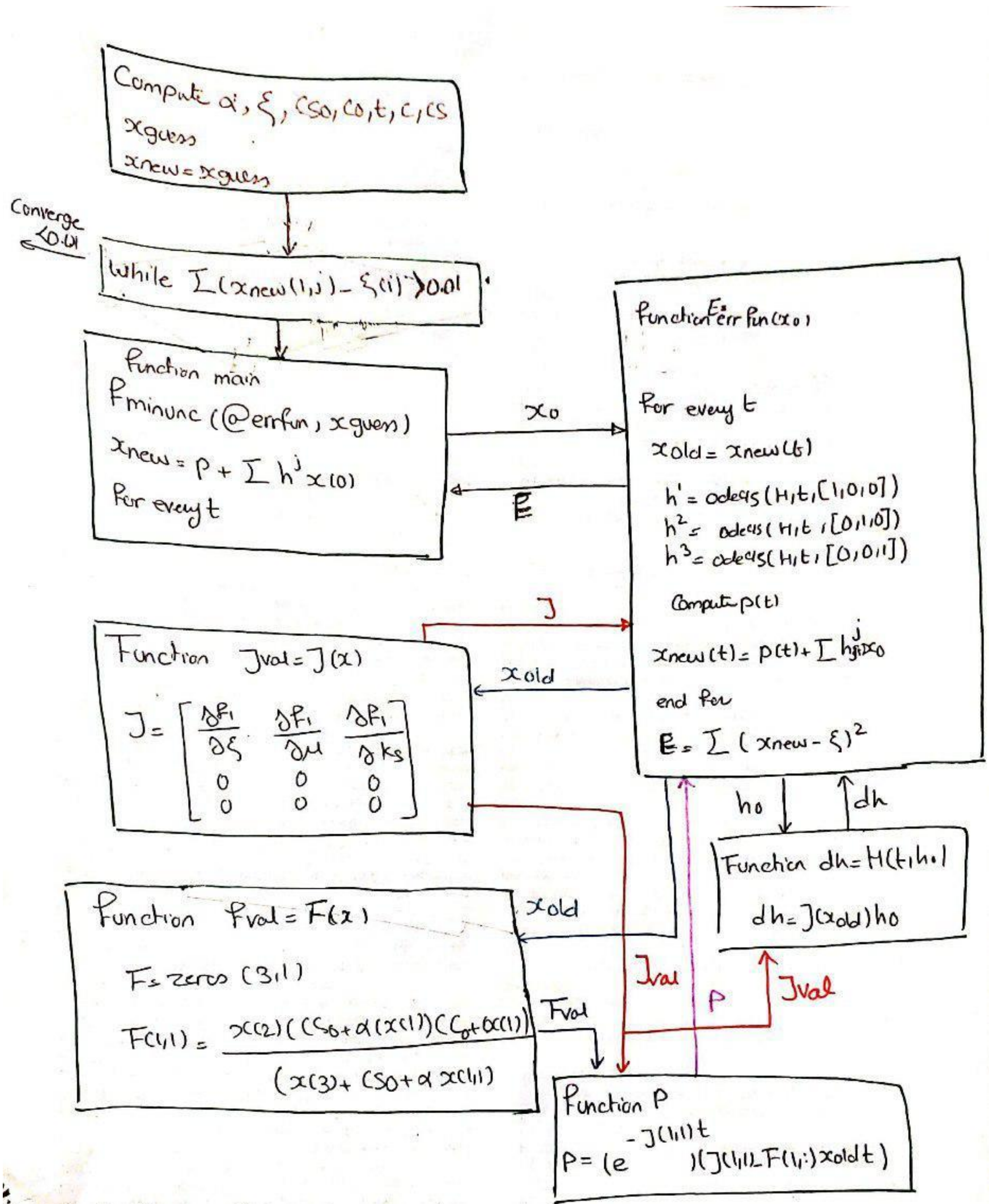


Define all your Functions
In the main Function